



Can artificial intelligence understand our emotions? Deep learning applications with face recognition

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Abstract

Objective The aim of this study is to evaluate the ability to detect emotions from human facial expressions via facial recognition technologies and analyze the effectiveness of deep learning models in this process.

Method This research was conducted between 01.04 and 01.07.2024. The data of the study were taken from the open access site <https://www.kaggle.com/datasets/msambare/fer2013> (Kaggle, 2024). Python 3.8 is used in this study. The FER-2013 (Facial Expression Recognition 2013) dataset is a comprehensive collection of facial images labeled with various emotions. The dataset contains 35,887 grayscale facial images. Each image has a size of 48×48 pixels. The dataset consists of images belonging to 7 emotion categories: anger, disgust, fear, happy, sad, confused, and neutral.

Results In our experiments on the FER2013 dataset, we evaluated the performance of three different models: MobileNetV3-L, EfficientNetV2-L, and our proposed EfficientMobileNet. The evaluation criteria were based on sensitivity, specificity, accuracy, and F1 scores to assess the effectiveness of each model comprehensively. The EfficientMobileNet model outperformed MobileNetV3-L and EfficientNetV2-L in all measured performance metrics. EfficientMobileNet was the most successful model for predicting emotions, with an accuracy of 77.6%.

Conclusion The impressive results obtained by EfficientMobileNet on the Fer2013 dataset show potential for wider application, especially in image classification tasks involving low-quality or small-scale images. This performance supports the idea of the potential for further improvements in neural network architecture and model efficiency and accuracy. Future work should focus on optimizing the model for more challenging datasets, studying the impact of different architectural adjustments, and investigating the scalability of EfficientMobileNet across various domains and applications.

Keywords Artificial Intelligence · Emotions · Deep learning

Introduction

Artificial intelligence (AI) encompasses the study and application of knowledge to develop systems that replicate human intellectual processes. By harnessing computing power,

AI aims to emulate human behaviors and equip machines with the ability to learn, reason, and make decisions akin to human beings (Zhang & Lu, 2021). Artificial intelligence, also known as machine intelligence, is the technological capability that allows computer systems to interpret and

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learn from data (Chen, M., & Decary, M., 2020). Unlike traditional software, AI systems are designed to analyze patterns, make predictions, and draw conclusions from past interactions or historical data, thereby mimicking the cognitive functions typically associated with human minds, such as problem-solving and decision-making (Gupta et al., 2021). Artificial intelligence (AI) integrates elements from multiple disciplines, including computer science, logic, biology, psychology, and philosophy. This interdisciplinary approach has led to significant advancements in various applications (Abonamah et al., 2021). These include speech recognition, image processing, natural language processing (NLP), automatic theorem proving, and the development of intelligent robots. The applications of AI shows its ability to mimic and enhance human cognitive functions (Zhang & Lu, 2021). Facial emotions play a crucial role in human communication, assisting in the interpretation of others' intentions. Typically, individuals rely on facial expressions to gauge emotional states such as joy, sadness, and anger (Ahmad et al., 2023). Research suggests that while verbal communication accounts for approximately one-third of human interaction, nonverbal cues constitute two-thirds (Ko, 2018). Among these nonverbal cues, facial expressions are particularly significant, serving as vital conduits for emotional and interpersonal communication. Given their importance, it is unsurprising that research into facial expressions has gained considerable interest in recent years (Kaulard et al., 2012). This field of study not only is pertinent to the perceptual and cognitive sciences but also has found applications in emotional computing and computer animation, where understanding and replicating human expressions are key (Ko, 2018).

The human face serves as a crucial biometric tool for identification, commonly known as 'face recognition' (Van Dyck & Gruber, 2023). This technology has become increasingly prevalent on a commercial scale and is integrated into a wide array of devices. From smartphones to home security systems, facial recognition uses unique facial features to verify the identities of individuals, offering convenience and security in everyday technology (Aguileta et al., 2022). Another potentially very useful subfield of facial analysis is emotion recognition via facial expressions (Saadon et al., 2023). Ekman categorized facial emotions into seven basic categories: happiness, fear, sadness, surprise, anger, contempt, and disgust. These categories form the core of his widely recognized model in the study of facial expressions, which posits that these emotions are universal across different cultures (Ekman, 1992). Emotion is an important human trait that affects the physical and mental state of a person (Chen et al., 2021). Emotional analysis is concerned with the measurement and classification of emotional states in individuals and is useful for inference-related tasks such

as perception, recognition, recommendation, monitoring, and prediction. Emotion analysis is promising for learning science and digital health applications (Tan et al., 2023). Currently, methods for emotion analysis are primarily categorized into two types: dictionary-based and machine-learning-based methods. Dictionary-based approaches utilize sentiment dictionaries, predefined rules, and sentiment functions as external sources to analyze and determine the emotional tone of a text (Hsieh & Zeng, 2022). In contrast, machine-learning-based methods treat emotion analysis as a classification problem. These approaches employ advanced techniques, including machine learning algorithms and deep neural networks, to classify text accurately into specific sentiment categories. Each method offers unique advantages depending on the complexity of the text and the precision of the analysis (Khanbhai et al., 2021).

Machine learning (ML) systems are designed to automatically learn models from data, enhancing their decision-making capabilities over time. Deep learning, a subset of machine learning, has become a dominant approach because of its effectiveness in handling large volumes of data and its success in various computer vision tasks (Choi et al., 2020). Among the deep learning architectures, convolutional neural networks (CNNs) are particularly notable. CNNs operate by processing input images through multiple layers of filters in convolutional layers, which help in extracting and learning features by generating a series of feature maps (Coleman et al., 2022). These feature maps are then integrated into fully connected layers that, in turn, utilize the Softmax algorithm to classify facial expressions into predefined categories. This process allows CNNs to achieve state-of-the-art results in recognizing complex patterns such as facial expressions (Ko, 2018). CNNs take images as inputs and are capable of distinguishing between various types of images via learnable weights and biases. This makes them particularly effective for image recognition, classification, and analysis (Saeed et al., 2022).

Facial expression recognition presents a complex challenge in the field of artificial intelligence (Liu et al., 2023). Achieving a comprehensive understanding and accurate prediction of emotional cues demands the rigorous integration of multiple contextual factors. Furthermore, it is essential for these systems to operate effectively in real-world environments to ensure that predictions are accurate and that the emotional experiences created are realistic (Monica et al., 2023). To address these complexities, specialized facial expression recognition (FER) algorithms have been developed. These algorithms utilize analytical techniques to systematically interpret facial expressions to evaluate the emotional states of individuals (Barrett et al., 2019).

In this study, our objective was to analyze emotions through facial expressions. Accurately identifying and

categorizing facial expressions are key components of nonverbal communication and are essential for human interaction. By observing and correctly interpreting these expressions, individuals can share and understand subjective emotional states, fostering empathy and communication. This ability also serves as a critical reference for decoding the intentions behind another person's actions. Our research seeks to enhance the understanding of these processes, contributing to more effective and nuanced human interactions.

Method

Dataset

The data of the study were taken from the open access website <https://www.kaggle.com/datasets/msambare/fer2013> (Kaggle, 2024). The FER-2013 (Facial Expression Recognition 2013) dataset is a comprehensive collection of facial images labeled with various emotions. The dataset contains 35,887 grayscale facial images. Each image has a size of 48×48 pixels. The dataset consists of images belonging to 7 emotion categories: anger, disgust, fear, happy, sad, confused, and neutral. When training and evaluating deep learning models, it is important that the classes in the test data are balanced so that the model can recognize all classes equally well. Imbalances between classes may cause the model to overfit the dominant classes and increase the misclassification rate of underrepresented classes. In our study, we paid attention to balancing the classes in the test data.

The dataset is divided into three parts:

- - Training set: 28,709 images.
- - General test set: 3,589 images.
- - Special test set: 3,589 images.

Deep learning models

Deep learning is a form of machine learning that uses large neural networks to represent complex patterns and relationships in data. Deep learning networks, unlike neural networks used in classical machine learning, include many hidden layers that allow them to automatically learn complicated, hierarchical data representations (Goodfellow et al., 2016). This depth allows deep learning models to detect complicated patterns and nonlinear correlations that simpler networks cannot. This makes them especially useful for unstructured data, including images, text, and audio. One of the most important deep learning architectures is the CNNs which are ideal for image and spatial data. Their unique structure enables convolutional layers to automatically extract important features from the data.

Convolutional neural network (CNN)

CNN is a class of deep neural networks commonly used to analyze visual data. A CNN model consists of a number of layers, such as convolutional layers, pooling layers, and fully connected layers (Krizhevsky et al., 2012).

Convolutional layers

These layers apply convolutional operations to the input image via a set of learnable filters (kernels). Each filter convolves along the width and height of the input, producing a 2D feature map representing the activation of the filter at each spatial location.

Activation function

After each convolutional operation, an activation function such as ReLU or tanh is applied to the model to linearize the Nonlinearity.

Pooling layers

Pooling layers play a critical role in convolutional neural networks (CNNs) by reducing the computational load and helping prevent overfitting. They achieve this by decreasing the spatial dimensions—specifically, the width and height—of the input volume as it passes through the network. The most commonly used type of pooling is maximum pooling. This method involves selecting the maximum value from each patch of the feature map. By doing so, maximum pooling effectively reduces the amount of data, focusing on the most prominent features, and simplifies the information that the network needs to process in subsequent layers.

Fully connected layers

Fully connected layers, typically positioned at the end of a convolutional neural network (CNN), function similarly to traditional neural networks. In these layers, every neuron is interconnected with every neuron in the preceding layer. The activations of these neurons are computed via matrix multiplication, which is then adjusted by adding a bias offset. This comprehensive connectivity allows the network to integrate learned features from prior convolutional layers, aiding in the final decision-making process, such as classification or regression tasks.

Output layer

The output layer in many neural network models, particularly those used for classification tasks, is typically a

Softmax layer. This layer functions by converting the raw output scores, often referred to as logits, from the final fully connected layer into a probability distribution. Each value in this distribution represents the probability that the given input corresponds to one of the class labels. The Softmax layer ensures that the sum of these probabilities equals one, making it possible to interpret the output as a likelihood measure for each class. This characteristic makes the Softmax layer essential for models where a clear probabilistic interpretation is necessary for decision-making.

The general block diagram of a CNN algorithm is shown in Fig. 1.

MobilenetV3

The MobileNetV3 architecture is a lightweight convolutional neural network designed for mobile and edge devices (Howard et al., 2019). It builds on the efficiency of previous versions (MobileNetV1 and V2) by incorporating advanced techniques such as compression and excitation modules and the use of the h-swish activation function. The SE modules help improve the representational power of the network by adaptively recalibrating the feature responses on a channel-by-channel basis (Wang et al., 2023). It is integrated into various stages of the network. The h-swish activation function is a hardware-friendly approximation of swish activation and improves the accuracy and efficiency of the model. Like MobileNetV2, MobileNetV3 uses inverse residual blocks with linear bottlenecks, which helps to maintain representational power while reducing computational complexity.

EfficientNetV2

EfficientNetV2 uses a compound scaling method that simultaneously scales depth, width, and resolution to improve model performance (M. Tan & Le, 2021). M. Tan & Le used a composite scaling method that simultaneously scales depth, width, and resolution to improve model performance.

The architecture introduces Fused-MBConv blocks in early layers and MBConv blocks in later layers for better performance. Fused-MBConv combines depth-directional convolution and point-directional convolution into a single process, which is more efficient for early layers. Considering the training speed and accuracy when designing the architecture, a training-aware NAS was used, and the structure of the network was optimized for faster training. EfficientNetV2 is divided into several stages, each consisting of multiple layers of Fused-MBConv or MBConv blocks. It combines deeply separable convolutions and efficient building blocks to reduce the computational cost. The composite scaling method provides balanced scaling across dimensions for optimal performance.

Proposed model efficientmobilenet

EfficientMobileNet is a new model that was obtained by combining the strengths of EfficientNetV2 and MobileNetV3 architectures. Combining EfficientNetV2's compound scaling technique, MobileNetV3's Compression and Excitement (SE) modules, Fused-MBConv blocks, and h-swish activation function, it aims to reduce the computational cost and increase the accuracy.

Efficientmobilenet architecture

EfficientMobileNet consists of four main components: Fused-MBConv blocks, SE blocks, MBConv blocks, and container model architecture. Each component is designed to increase the efficiency and accuracy.

Fused-MBConv modules

In Fused-MBConv blocks, EfficientNetV2's "Fused-MBConv" method extracts better features than traditional MBConv blocks. This block consists of two main stages:

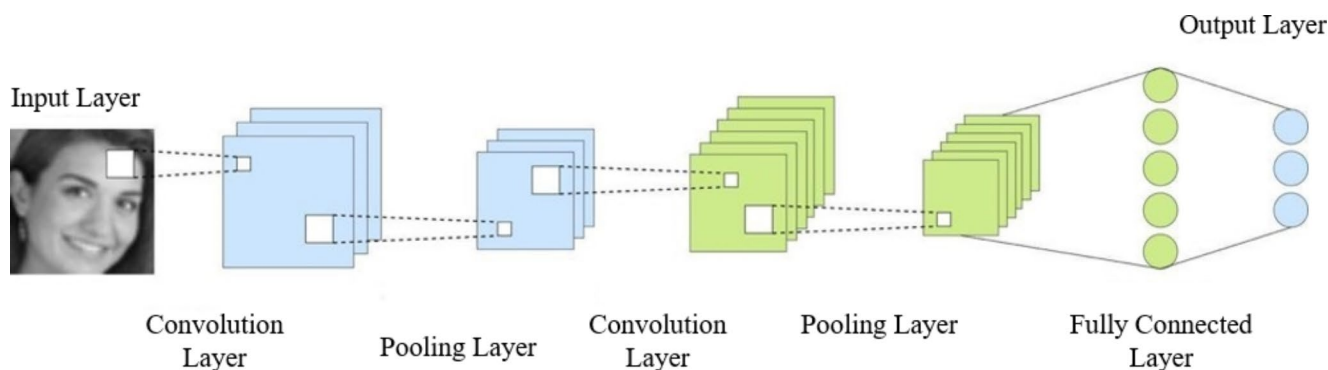


Fig. 1 CNN algorithm general block structure

Conv2D (1×1) Dilation Stage In the first stage, the input filters are increased by a certain dilation factor. This stage enables the model to obtain additional information by analyzing the data in multiple channels. The convolution layer allows the model to obtain more robust features by increasing the filters.

Depth Convolution Stage Deep discrete convolution (Deep Conv2D) is added to the developed feature map with a certain kernel size and step value. Deep convolution aims to significantly reduce the computational load by performing independent operations between channels. The BatchNormalization layer and the ReLU activation function normalize the output and include nonlinear features. Finally, the 1×1 Conv2D layer is used to reduce the number of filters to the initial level, and an output similar to the block input but enhanced with deeper features is obtained.

SE blocks (squeeze-and-Excitation)

As an essential feature of EfficientMobileNet architecture, SE blocks include a channel-based attention mechanism that assigns data weights according to the state of each channel. SE blocks ensure that important information in the feature map stands out by creating attention coefficients for each channel.

GlobalAveragePooling2D This process calculates the average of the input for each channel and gives a summary value for each channel. Global average pooling reduces the channels for the next layers by collecting and concentrating the comprehensive information from each channel.

Compression and Reshaping The results are concentrated according to a sparsity ratio (`se_ratio`) and reordered for each channel. This stage provides more efficient data processing and reduces the computational load by reducing the number of channels.

ReLU activation with Conv2D (1×1) ReLU activation is applied per channel for compressed data. This stage includes an activation function that determines the attention level assigned to the channels.

Conv2D with sigmoid activation (1×1) Attention coefficients for each channel are generated using the sigmoid activation function. The coefficients are multiplied with the input data using the Multiply operation, and the attention

coefficients of each channel are summed. This looks at the state of each channel in the model.

MBConv blocks

EfficientMobileNet includes traditional MBConv blocks. These blocks are designed on top of previous models' basic components, including EfficientNet and MobileNet. MBConv blocks help identify deeper features by improving channel-based information processing during feature extraction.

Expansion phase Input filters are increased by an expansion factor and processed with additional channels. In this expansion phase, the H-swish or ReLU activation function is used. The expanded output is subjected to Deep Convolution. The process then proceeds using the H-swish or ReLU activation function. Deep convolution facilitates more complex feature extraction by increasing computational power.

SE Blocks In this step, SE blocks apply the attention mechanism channel-by-channel basis, emphasizing more salient data features. **Projection Step:** In this final step, the filter amount is returned to the input level and reduced to the initial number of channels. This step ensures that the data from the block is consistent with the input.

General framework of efficientmobilenet model

EfficientNetV2 and MobileNetV3 are inspired by EfficientMobileNet architecture to improve efficiency and accuracy. The model has several aspects within its overarching framework:

Input layer The first layer extracts the input features of the model through Fused-MBConv blocks. These blocks optimize the data processing efficiency and provide high information density in the first stage of the model.

Deep layers Feature extraction of data is performed using consecutive MBConv blocks and SE blocks in deeper layers of the model. The H-swish activation mechanism is incorporated in EfficientMobileNet, similar to MobileNetV3, and is designed to offer advantages, especially for mobile devices.

Last layers In the last stage of the model, a summary feature vector is obtained using the GlobalAveragePooling2D layer. Then, classification is performed using a fully connected

layer, and the class prediction is achieved by softmax activation in the last layer.

The EfficientMobileNet architecture meets the contemporary image classification requirements by combining the composite scaling and Fused-MBConv blocks of EfficientNetV2 with the SE modules and h-swish activation mechanism of MobileNetV3. It is an excellent choice for small-scale mobile devices and large-scale server applications, offering an outstanding balance between efficiency and precision. While the ReLU activation function is preferred due to its simplicity and efficiency, the h-swish activation provides a more sophisticated result, especially in mobile and resource-constrained environments. EfficientMobileNet addresses the contemporary classification problems by integrating the composite scaling and Fused-MBConv blocks of EfficientNetV2 with the SE modules and the h-swish activation function of MobileNetV3, resulting in a versatile model suitable for many applications. The general structure of the EfficientMobileNet architecture is detailed in Fig. 2.

Results

In our experiments on the FER2013 dataset, we evaluated the performance of three different models: MobileNetV3-L, EfficientNetV2-L, and our proposed EfficientMobileNet. The evaluation metrics included sensitivity, specificity, accuracy, and F1 scores to assess the effectiveness of each model comprehensively.

Sensitivity measures the model's ability to identify true positives accurately. It shows how well the model detects instances of the target class. Specificity measures the model's ability to correctly identify true negatives. It reflects how well the model avoids false positives. The accuracy represents the overall accuracy of the model and is calculated as the ratio of correctly predicted samples to total samples. The F1 score is the harmonic mean of precision and recall, providing a metric that balances both concerns and is especially useful for imbalanced datasets. To better understand the model behaviors, we analyzed the complexity matrix, which provides detailed insights into the classification results, highlighting true positives, false positives, true negatives, and false negatives for each class. A transfer learning structure was used to train the models. The ImageNet weights were given as inputs to the training for all 3 models as initial weights. The training and validation accuracy graphs for the training of the EfficientMobileNet model are shown in Fig. 3. The model demonstrates a strong learning capability, stabilizing near 75–80% validation accuracy, which reflects the model's effectiveness in capturing patterns in the data and achieving fair generalization. Although a slight gap exists between training and validation accuracy, the overall trend indicates the trained model performs well.

As shown in Table 1, the EfficientMobileNet model outperformed the MobileNetV3-L and EfficientNetV2-L models in all the measured performance metrics. This performance emphasizes the superior capability of the EfficientMobileNet architecture in performing image classification tasks. In particular, the model's performance on

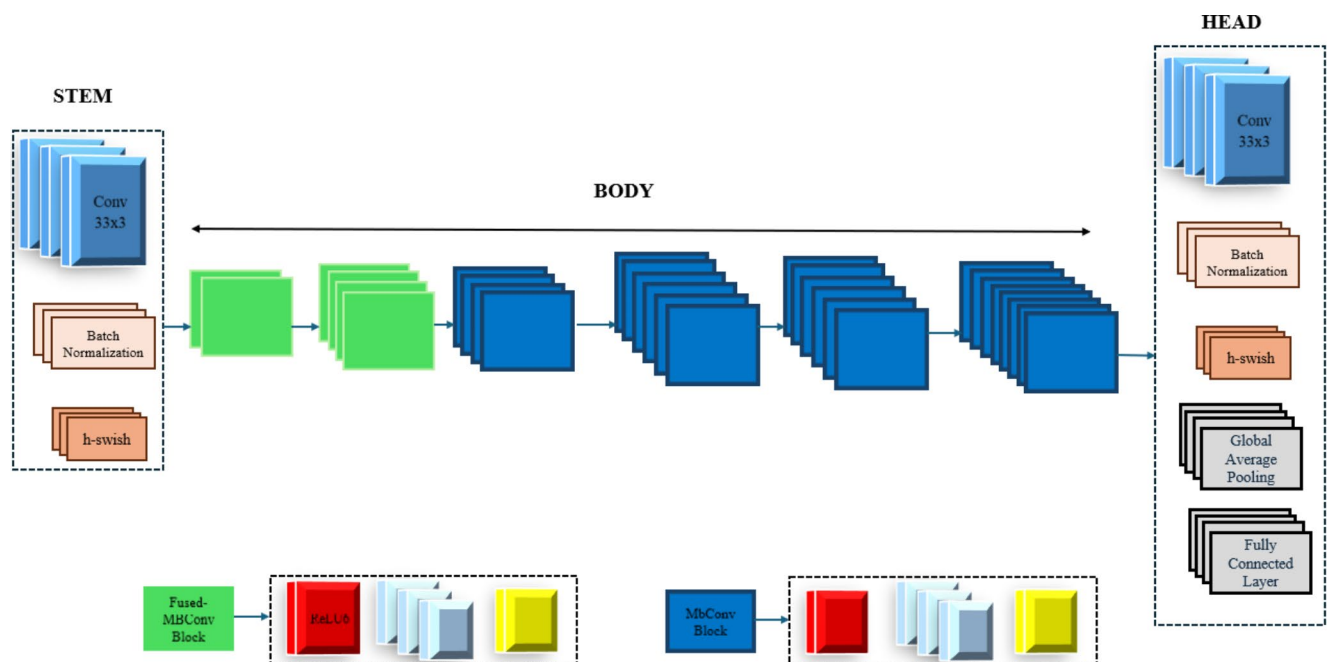


Fig. 2 The general structure of the layers of the EfficientMobileNet architecture

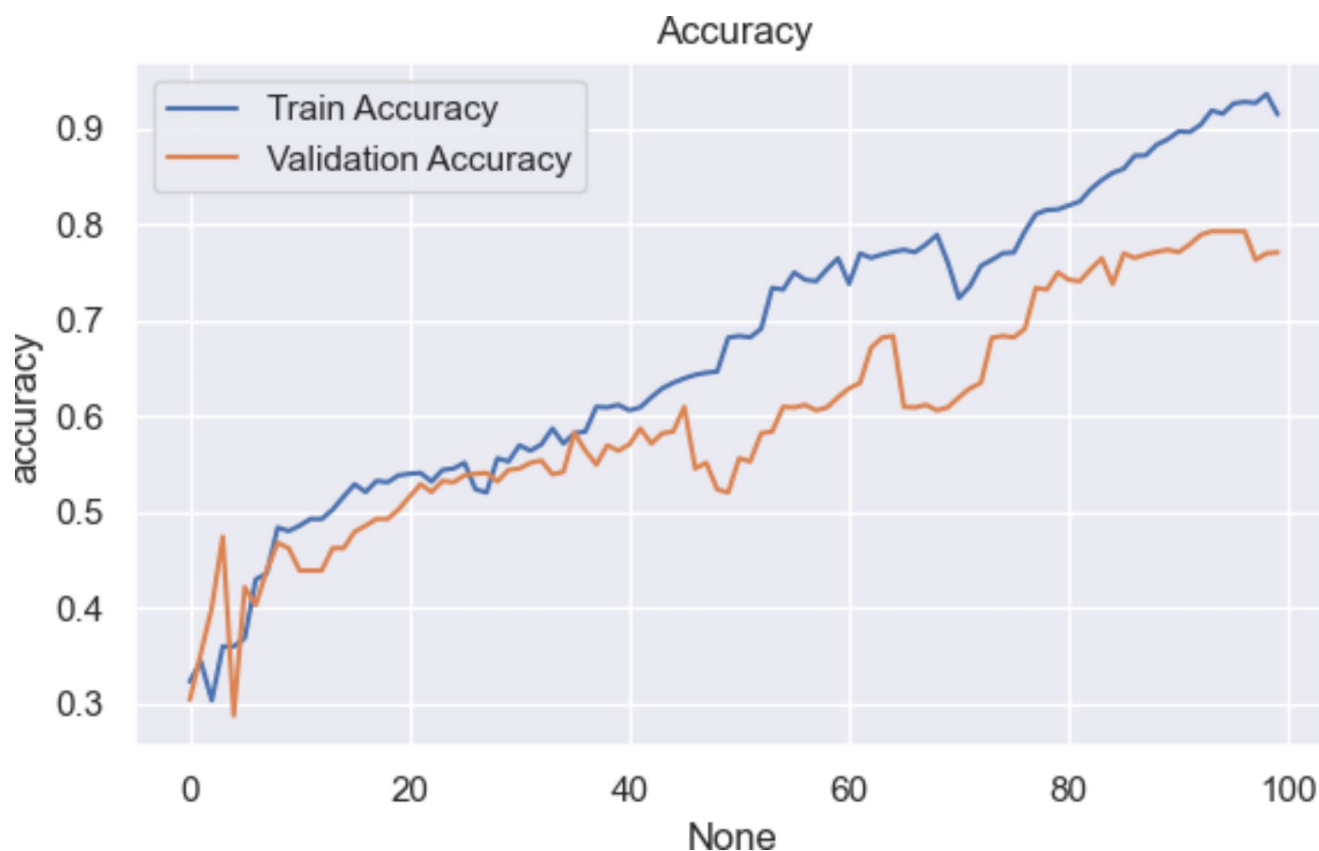


Fig. 3 EfficientMobileNet model training and validation accuracy graphs

Table 1 Overall performance results of the studied and proposed architectures on test data

Models	Precision(%)	Recall(%)	F1-score(%)	Accuracy(%)
Efficient-NetV2-L	69.1	68.7	69.1	70.4
Mobile-NetV3-L	73.3	69.5	71.2	71.6
EfficientMobileNet	77.1	77.3	76.8	77.6

different evaluation metrics demonstrates its robustness and reliability. The Fer2013 dataset, known for its small-scale images and relatively low image quality, poses a challenge for image classification models. Despite these challenges, the EfficientMobileNet model performs competitively.

Figure 4 shows the performance of the EfficientMobileNet model by class using the confusion matrix obtained from the test data. The model demonstrates strong performance in recognizing several emotions, especially happiness, anger, and fear, showing high accuracy and precision. Its ability to accurately categorize happiness nearly perfectly and distinguish anger effectively emphasizes its reliability with specific emotions.

The results obtained by EfficientMobileNet on the Fer2013 dataset show potential for wider application, especially in image classification tasks involving low-quality or small-scale images. This performance supports the idea of the potential for further improvements in neural network architecture and model efficiency and accuracy.

Discussion

In our experiments on the FER2013 dataset, we evaluate the performance of three different models: MobileNetV3-L, EfficientNetV2-L, and our proposed EfficientMobileNet. The evaluation criteria were based on sensitivity, specificity, accuracy, and F1 scores to assess the effectiveness of each model comprehensively. The EfficientMobileNet model outperformed MobileNetV3-L and EfficientNetV2-L in all measured performance metrics. EfficientMobileNet was the most successful model for predicting emotions, with an accuracy rate of 77.6%. The results in the literature are given in Table 2 (Chowdhury et al., 2024).

On the FER-2013 dataset, the CLCM achieved 63% accuracy, surpassing the results of TouchyFeely (Amin et al., 2017), MI+MII+MIII (Zeng et al., 2018), and EXPDAL-MSE (Zhai & Li, 2018) and equaling Tiny Xception

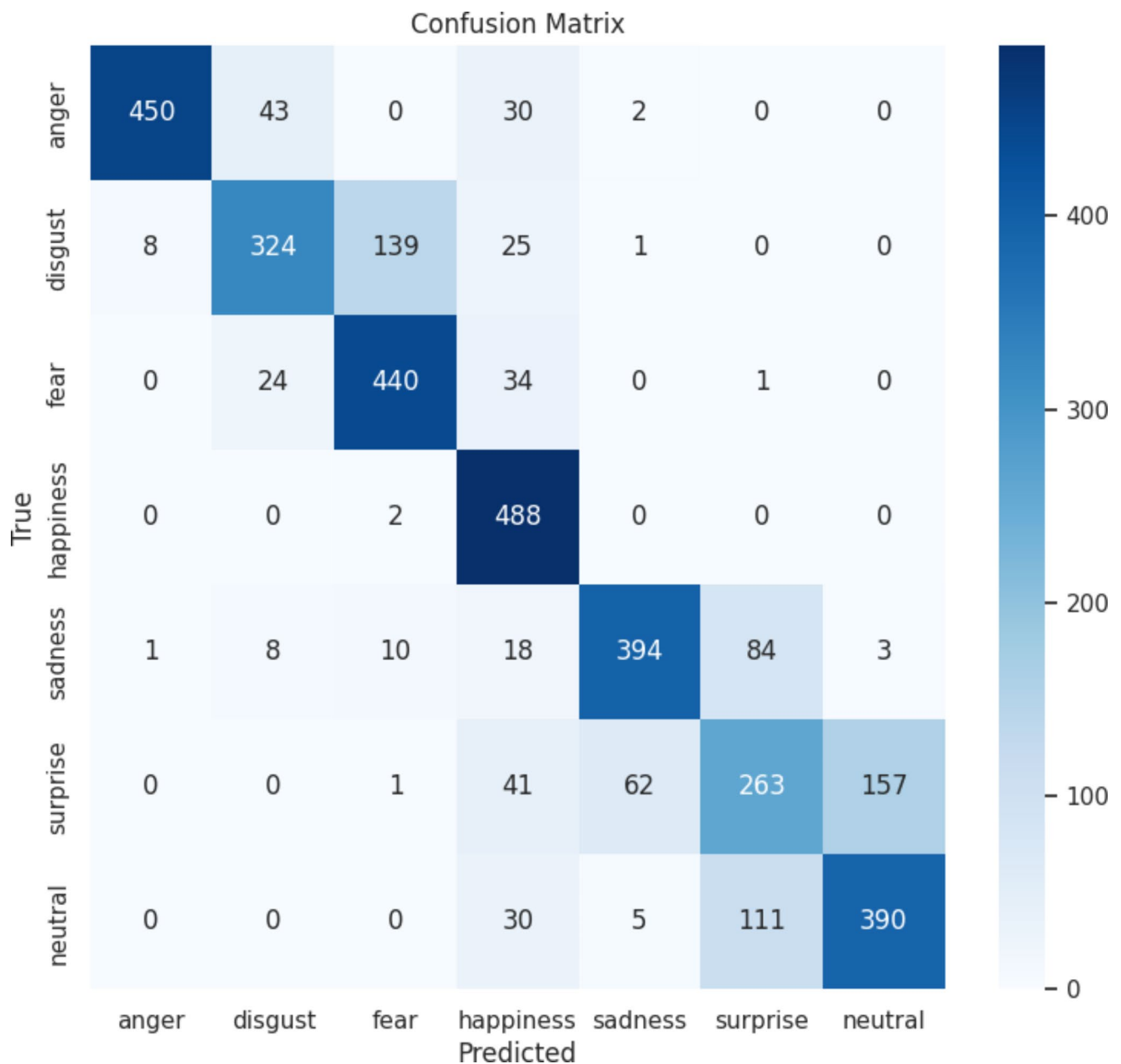


Fig. 4 Confusion matrix of the EfficientMobileNet model on test data

(Raksarikorn & Kangkachit, 2018). However, it was slightly outperformed by ShuffleNetV2, which recorded 65% accuracy. On the RAF-DB dataset, the CLCM demonstrated significant improvement with 83% accuracy, surpassing AlexNet (Li et al., 2017), VGG (Li et al., 2017), BaseDCNN (Li et al., 2017), MobileNetV2, and ShuffleNetV2. In terms of performance on the AffectNet dataset, the CLCM achieves a score of 0.54, which is comparable to or better than that of the 2att-CNN (Xiaohua et al., 2019), VGG-16 (Li et al., 2018), P-CNN (Li et al., 2018), and DLP-CNN (Li et al., 2018) but slightly lower than the 57% accuracy of MobileNetV2 and ShuffleNetV2. This suggests a potential

advantage for MobileNetV2 and ShuffleNetV2 in processing datasets with larger images. Notably, many studies in the literature have omitted confusion matrices, focusing solely on reporting overall accuracy.

Mozaffari et al. (2023) achieved 62% accuracy in their analyses using the FER-2013 dataset. Their model implemented batch normalization and did not balance image classes. Sarvakar et al. (Sarvakar et al., 2023) proposed a CNN model that achieved 54% accuracy, and these two models performed lower than the CLCM. Our study was found to yield more accurate predictions than those of all existing studies.

Table 2 Performance results of various model configurations on the FER2013 dataset

Model Configuration	Dataset	Accuracy (%)	Weighted-F1 (%)	AUC-ROC (%)	AUC-PRC (%)
VGG19 with all layers is frozen+No Histogram	FER2013	35.99	29.00	57.70	19.73
Fine-tuned VGG19+No Histogram	FER2013	69.06	68.57	80.87	52.26
Fine-tuned VGG19+Histogram	FER2013	69.44	68.34	80.20	51.58
VGG16 with all layers is frozen+No Histogram	FER2013	41.20	35.53	60.36	22.22
Fine-tuned VGG16+No Histogram	FER2013	68.80	69.29	81.34	52.37

Emotion analysis through facial recognition via deep learning has received significant attention in several studies. The application of deep learning techniques in facial emotion recognition has shown promising results for emotion detection and human-computer interaction (Xu, 2024). Researchers have explored combinations of different technologies, such as convolutional neural networks (CNNs) and local binary patterns, to improve facial emotion recognition, particularly in areas such as criminal investigations and psychological feature extraction (Zhu et al., 2022). Moreover, multimodal emotion recognition (MER) systems aim to integrate facial expression, speech, and electroencephalogram (EEG) data via deep learning approaches and improve the overall performance of emotion recognition systems (Pan et al., 2023). Studies have also focused on developing efficient models for facial emotion recognition via deep convolutional neural networks (DCNNs) and multiblock deep learning processes to analyze core facial areas for emotion detection (Shin et al., 2019). Furthermore, the design of blended instructional models based on emotion recognition and language learning highlights the potential of deep learning algorithms in achieving facial emotion recognition in educational settings (He & Liu, 2022). Researchers have also presented new approaches, such as deep facial expression vector extraction (DeepFEVER), to create a generic deep feature extractor for facial expression recognition tasks (Schoneveld & Othmani, 2021). In the context of human-robot interaction, studies have explored the use of gated repetitive units (GRUs) and robotic process automation for facial emotion recognition and demonstrated the application of deep learning techniques in improving robot-human interactions (Saxena et al., 2020). In addition, research efforts have been directed towards optimal emotion recognition via deep learning algorithms, which focus on detecting facial expressions, such as happiness and sadness,

through convolutional neural networks (Kumar Arora et al., 2022). Overall, the synthesis of these studies highlights the growing interest in leveraging deep learning methodologies for facial expression recognition, highlighting the potential for advancements in sentiment analysis, human-computer interaction, and psychological feature extraction.

Conclusions

The impressive results obtained by EfficientMobileNet on the Fer2013 dataset show potential for wider application, especially in image classification tasks involving low-quality or small-scale images. This performance supports the idea of the potential for further improvements in neural network architecture and model efficiency and accuracy. Future work should focus on optimizing the model for more challenging datasets, studying the impact of different architectural adjustments, and investigating the scalability of Efficient-MobileNet across various domains and applications.

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Data Availability The data can be obtained from Kaggle using the link below:
<https://www.kaggle.com/datasets/msambare/fer2013>.

Declarations

Conflicts of interest The authors report no actual or potential conflicts of interest.

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